

Sumalatha Panja

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Summary

Data Engineer with 4+ years of experience designing and implementing scalable, cloud-native data pipelines and analytics platforms. Skilled in real-time streaming (Kafka, Kinesis, Spark Structured Streaming), batch ETL (Databricks, Azure Data Factory, NiFi), and building ML-ready datasets for fraud detection, credit scoring, and predictive analytics. Proficient in Python, PySpark, SQL, dbt, Snowflake, Redshift, ClickHouse, Airflow, and modern data governance tools, delivering high-performance, compliant, and resilient solutions that drive operational and business decisions.

Technical Skills

- **Data Engineering & ETL** – Real-Time Streaming (Apache Kafka, AWS Kinesis, Spark Structured Streaming), Batch ETL (Azure Data Factory, Databricks, Nifi), Python, Pyspark, SQL, Pandas, Delta Lake, Change Data Capture (CDC), Talend, Matillion, Airbyte, Apache Flink.
- **Cloud & Data Warehousing** – AWS, Azure, Snowflake, Amazon Redshift, Azure Synapse Analytics, Clickhouse, Apache Iceberg, Azure Data Lake, Google Bigquery, AWS Glue, Redshift Spectrum, Databricks Lakehouse, Terraform, Cloud Cost Optimization.
- **Data Pipelines & Orchestration** – Airflow, Dbt, Databricks, Fastapi, Spark, Pyspark Structured Streaming, Prefect, Luigi, Jenkins, Kubernetes, Docker, Miflow, Event-Driven Architecture, REST API Integration.
- **Analytics, AI & ML Integration** – Feature Engineering, Predictive Modeling Pipelines, Demand Forecasting, Churn Prediction, Fraud Detection, Operational Intelligence Dashboards, Time-Series Analytics, Anomaly Detection, ML-Ready Datasets, Tensorflow, Pytorch, Scikit-Learn, Xgboost.
- **Data Governance & Observability** – Great Expectations, Monte Carlo, Openlineage, Apache Atlas, Automated Validation, PCI DSS And HIPAA Compliance, Data Quality Monitoring, Data Cataloging, Metadata Management, GDPR Compliance, RBAC, Schema Evolution, Data Lineage Tracking.
- **Tools & Technologies** – Git, JIRA, REST Apis, Mysql, Postgresql, Hive, HDFS, Iot/Telemetry Data Integration, Real-Time Alerts, SLA Monitoring, Snowsql, Dbt Cloud, Tableau, Power BI, Looker, JSON, Parquet, Avro, Linux/Unix Scripting, Bash, CI/CD Pipelines, Agile/Scrum Methodologies.

Professional Experience

Data Engineer, American Express, Inc.

07/2024 – Present | Remote, USA

- Designed an AI-driven real-time credit analytics platform integrating transactional, payment and cardholder data with risk and underwriting teams, enabling event-driven risk scoring, SLA monitoring and governance, improving credit decision efficiency by 16%.
- Crafted a resilient low-latency streaming pipelines using Apache Kafka and AWS Kinesis to ingest card transactions, fraud alerts and authorization events. Hybrid approach handles structured transaction logs and semi-structured event JSONs, reducing fraud detection latency by 20%.
- Developed scalable data services with Python, FastAPI and PySpark to enrich, validate and serve financial datasets powering credit dashboards, alerts and operational intelligence. Designed data schema, staging and presentation layers and ETL orchestration.
- Architected cloud-native data layers using Snowflake, Amazon Redshift, Apache Iceberg and ClickHouse enabling sub-second OLAP, time-series transaction analytics and high-concurrency reporting, improving performance during peak cardholder activity by 18%.
- Orchestrated AI-ready data pipelines using Airflow, Databricks, Spark and dbt in collaboration with the Data Science team to generate governed feature datasets for credit scoring, churn prediction and fraud models, improving automation and accuracy by 15%.
- Strengthened data quality, lineage, security and compliance with Great Expectations, Monte Carlo, OpenLineage, DataHub, encryption and access controls, ensuring PCI DSS and HIPAA-aligned governance for sensitive cardholder and transaction data enterprise-wide.

Data & ETL Engineer, Uber India

03/2021 – 07/2023 | Hyderabad, India

- Engineered end-to-end trip, rider, driver and logistics telemetry pipelines integrating GPS, payments and operations data into raw, curated and presentation layers, improving ETA accuracy, ride allocation efficiency and operational decision-making by 45% across Uber India.
- Planned a scalable batch and streaming ETL pipelines using Azure Data Factory (ADF), Apache Kafka and Spark Structured Streaming, ingesting data from MySQL, PostgreSQL, GPS APIs and driver devices into HDFS, Hive and Azure Data Lake, reducing latency 65%.
- Built real-time ride analytics pipelines using Kafka and PySpark Structured Streaming, creating trip and driver schemas for anomaly detection, surge pricing and predictive allocation, improving on-time performance, fraud detection and customer satisfaction by 30%.
- Optimized large-scale ETL workloads on Databricks and Azure Synapse Analytics, processing terabytes of ride, payment and driver telemetry data daily, reducing processing time 70% while lowering cloud compute costs and improving operational efficiency.
- Settled a transformation and enrichment frameworks with Python, Pandas and SQL to unify trip, payment and user behavior schemas, ensuring consistency across raw, curated and presentation layers, supporting finance, operations and regional performance dashboards.
- Implemented ML-ready pipelines for demand forecasting, dynamic pricing and route optimization using Azure Databricks Lakehouse, improving predictive surge accuracy, trip fulfillment rates and driver engagement across urban and semi-urban regions of India by 25%.
- Strengthened data governance, observability and reliability using Great Expectations, Apache Atlas and ADF automated validation, improving data quality, schema compliance and audit readiness across Uber India's trip, driver and payment datasets by 50%.

Education

California State University, San Bernardino – San Bernardino, CA

Master of Science (MS) in Information Systems & Technology, Business Intelligence and Analytics

08/2023 – 05/2025

Institute of Aeronautical Engineering – Telangana, India

Bachelor of Technology (B.Tech) in Electrical and Electronics Engineering

07/2018 – 05/2022

Publication

Usage and Engagement Patterns of Students in Online Learning Platforms

- Analysis of student enrollments, course completion rates, and engagement metrics across multiple online learning platforms revealed that edX has the highest engagement, with students more likely to complete courses, indicating its effectiveness and appeal.
- Students prefer platforms offering courses with lower cost and shorter duration, showing that affordability and time efficiency strongly influence enrollment decisions, course completion, and overall engagement, highlighting trends for optimizing online learning experiences.